Practical – 14

Practical Title: Admin Dashboard with Role-Based Access Control, Dynamic Menu, and Audit Log

Problem Definition: Develop an admin dashboard that displays user statistics, student count, event count and latest registrations. Restrict access to admin users only and show dynamic menu options based on role.

A. Theory

1. Are dashboard values loaded dynamically from the database?
Yes. Dashboard values are loaded dynamically from the MySQL database using SQL COUNT() queries. The dashboard displays the total number of users, students, events, and registrations. The latest registrations are also retrieved using SQL JOIN queries.

2. Is access restricted to admin users?
Yes. Access is restricted using PHP sessions and the requireAdmin() function. It checks whether the user is logged in and has the admin role. Non-admin users are denied access to the dashboard.

3. Are role-based menus working?
Yes. Menu options are generated dynamically according to the logged-in user's role. The admin role receives menu options such as Dashboard, Users, Students, Events, Registrations, and Audit Logs.

4. Are important admin actions recorded in audit_logs?
Yes. Important actions, such as admin login, admin logout, and denied access attempts by non-admin users, are recorded in the audit_logs table along with the user ID and timestamp.

B. Implementation/ Output/Result:

The Admin Dashboard with Role-Based Access Control, Dynamic Menu, and Audit Log was developed using PHP, MySQL, HTML, and MySQLi. A MySQL database named practical14_db was created with five tables: users, students, events, registrations, and audit_logs to store and manage the required information.

PHP was used to establish database connectivity and execute SQL queries for retrieving and displaying dashboard information. A secure login system was implemented using PHP sessions and password verification to authenticate users. Access to the dashboard was restricted to users with the administrator role using a role-checking function.

The dashboard dynamically displays important statistics, including the total number of users, students, events, and registrations, using SQL aggregate functions. The latest student registrations are retrieved through JOIN queries and displayed in a table. A role-based dynamic menu was also implemented to display relevant navigation options according to the logged-in user's role.

Additionally, an audit logging mechanism was developed to record important activities such as admin login, logout, and unauthorized access attempts in the audit_logs table along with their timestamps. Finally, the application was tested by checking admin login, invalid credentials, non-admin access restrictions, dynamic dashboard statistics, and audit log entries.

C. Evidence of Work (Screenshots):

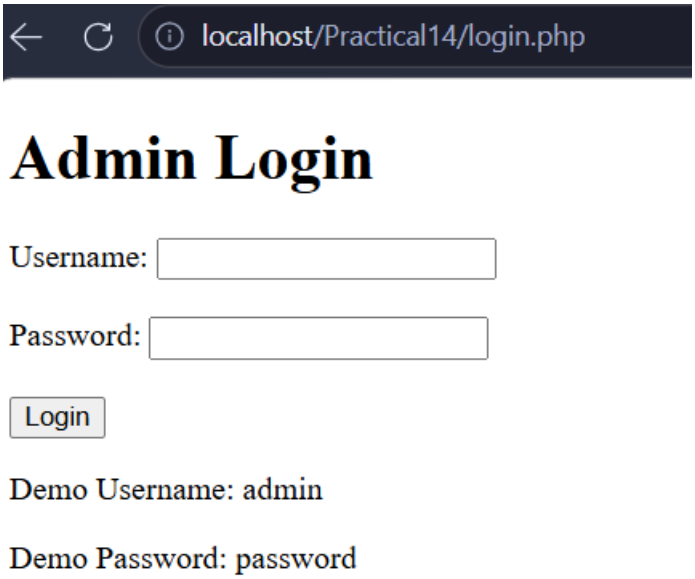


Fig1: Admin Login Page

Admin Dashboard

Welcome, admin

Role: admin

Dynamic Menu

Dashboard | Users | Students | Events | Registrations | Audit Logs | Logout

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Dashboard Statistics

Total Users: 3

Total Students: 3

Total Events: 3

Total Registrations: 3

Fig2: Admin Dashboard

Latest Registrations

ID	Student Name	Event Name	Registration Date
3	Neha Shah	Project Exhibition	2026-10-20 23:32:54
2	Rahul Patel	Coding Competition	2026-10-20 23:32:54
1	Aryan Chauhan	Web Development Workshop	2026-10-20 23:32:54

Fig3: Latest Registrations

Audit Logs

Username	Activity	Date and Time
admin	Admin logged in	2026-10-20 23:42:27
admin	Admin logged out	2026-10-20 23:41:53
admin	Admin logged in	2026-10-20 23:35:16

Fig4: Audit Logs

Admin Login

Invalid username or password.

Fig5: Access Restriction

D. GitHub Repository Link:

aryan150806-wq/StudentHubPortal: Web Development Framework P

E. Challenges Faced:

1. Establishing proper connectivity between PHP and the MySQL database and retrieving data from multiple related tables.
2. Implementing secure user authentication using PHP sessions and restricting dashboard access to administrators only.
3. Creating a dynamic menu based on user roles and ensuring that unauthorized users cannot access restricted sections.
4. Implementing the audit logging mechanism to record important activities such as login, logout, and denied access attempts.

F. Learning Outcomes:

The Admin Dashboard with Role-Based Access Control was successfully developed using PHP and MySQL. It displays dashboard statistics and recent registrations dynamically from the database. Role-based access control ensures that only authorized administrators can access the dashboard. The audit logging feature records important administrative activities for monitoring and tracking. Overall, this practical helped in understanding database connectivity, session management, role-based access control, and audit logging.

Faculty Remarks: